

A Typed Certification Protocol for Finite-State Causal Abstractions

The Finite Core of the DCS Research Program

Rongjie Wei

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Coarse-grained dynamics can be well defined while failing to preserve the targets or persistence events that motivate a model. This paper presents a typed certification protocol for finite-state, discrete-time controlled Markov systems. The protocol separates state aggregation, action implementation, history-response equivalence, and event-order quotients, and records their assumptions within a fixed task contract. For exact state abstractions, action-wise strong lumpability and implementation invariance yield well-defined macro kernels and finite-horizon state-path consistency under compatible control protocols. Probabilities of persistence events are preserved only when the events are measurable through the aggregated paths. For approximate abstractions, standard total-variation contraction and perturbation arguments provide a finite-horizon error budget with a separate term for target approximation. Finite examples and counterexamples distinguish dynamical closure from implementation consistency, target measurability, and preservation of identity-dependent events. The contribution is an explicit organization of these certification obligations and their failure witnesses, rather than new foundations for lumpability or Markov contraction. The results concern specified finite models and do not establish empirical validity in natural systems or general compositional guarantees for open systems. The protocol forms a bounded mathematical component of the DCS research program.

1 Declarations

- **Code and data availability:** The source code, fixed model configurations, expected outputs, tests, and reproduction instructions for this version are available at <https://github.com/weijierry/dcs-finite-certification> (public release, commit 2c15258, 2026-09-10). The release manifest lists the files and SHA256 checksums. All reported numerical examples are synthetic finite-model computations. A permanent archive record with a DOI is being registered.
- **Use of generative AI and AI-assisted tools:** Generative AI tools (GPT, Gemini, DeepSeek, and Doubao) were used during preparation of this manuscript for mathematical formulation assistance, drafting and revision of the exposition, reference organization, and consistency checking. All AI-assisted outputs were reviewed and verified by the author, who takes full responsibility for the content. AI tools are not authors, and their outputs are not treated as independent verification or scientific evidence.

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- **Correspondence:** weirongjie@f.mindas.me (Rongjie Wei; ORCID 0009-0002-0501-5570).

2 Introduction

2.1 The multi-scale object qualification problem

Many fields of modern science face the same problem: a system is described at the micro level by a large number of states and events, while research, prediction, and control must be carried out at some coarser macro level. The question is twofold. First, which macro objects are qualified to be treated as “the same object” persisting, being intervened upon, and being predicted? Second, which macro laws are qualified to be called “the laws of that system”?

The difficulty is not that coarse models are impossible. Coarse-grained descriptions are ubiquitous, and many of them are internally well defined. The difficulty is that being well defined as a dynamical system does not imply being faithful to the quantities that motivated the model. A macro kernel can be closed and stochastic while a response target defined on micro states is not measurable through the aggregation; a history compression can preserve the predictive distribution while losing intervention semantics; a partition can satisfy state lumpability while destroying the event poset; and persistence events that depend on micro identity may not be expressible at the macro level at all. This paper treats that gap as the object of study: we ask which certification obligations must be checked, separately and explicitly, before a macro description is used for a given target under a given task.

2.2 What established results already solve

Existing answers come from several independent traditions, each of which solves one facet of the problem:

- **Finite posets and causal sets** answer the structural question of whether one event can reach another (Bombelli et al. 1987; Norris 1997);
- **Graph rewriting** answers the generative question of which local transformations are lawful;
- **Finite conservative generators and stochastic semigroups** answer the dynamical question of how states evolve, in discrete and continuous time (Norris 1997; Kemeny and Snell 1976; Anderson 1991);
- **Strong lumpability** answers the question of which partitions yield autonomous macro Markov dynamics (Kemeny and Snell 1976; Buchholz 1994);
- **Causal consistency and causal abstraction** answer when macro and micro interventions commute (Rubenstein et al. 2017; Beckers and Halpern 2019);
- **Computational mechanics, information theory, and the information bottleneck** answer what the minimal predictively sufficient state is, and how much information coarse graining discards (Crutchfield and Young 1989; Shalizi and Crutchfield 2001; Tishby et al. 1999; Cover and Thomas 2006);

- **Mori-Zwanzig projection methods** answer how eliminated degrees of freedom generate memory (Zwanzig 1961; Mori 1965);
- **Finite-horizon coarse-graining error analysis** bounds the error introduced by coarse-grained hidden Markov dynamics (Andrieux 2012).

These results are mature mathematics. This paper does not claim originality for them; it claims that they do not, on their own, provide a unified typed contract in which structural, dynamical, response, intervention, memory, and persistence checks are registered separately and can fail separately.

2.3 The missing joint typed contract

The tools above each hold in their own domain, but “the same macro interface passing structural, dynamical, response, intervention, memory, and persistence checks simultaneously” lacks a unified form. Concretely:

- a partition may satisfy state lumpability yet destroy the event poset;
- a history compression may preserve the predictive distribution yet lose intervention semantics;
- a macro kernel may be dynamically closed yet be non-measurable for persistence events;
- an implementation correspondence may be well defined as a relation yet fail to be invariant across micro implementations.

The entry point of this paper is to register these checks as certification axes of one task contract, typed so as not to be confused with one another, rather than to build a single untyped map that does everything. Events, states, histories, and actions are placed in separate lanes; a single untyped arrow is forbidden from discharging two duties at once. The protocol then asks, for each axis, whether the corresponding premise holds, and pairs each dropped premise with a finite failure witness.

2.4 Contributions

The contribution of this paper is an explicit, task-specific certification protocol, not a new universal law:

1. **A typed decomposition of certification obligations.** State aggregation (per-action strong blockability, D5), action implementation (multi-implementation invariance, D6), target measurability (D7), and persistence-event measurability (D8) are separate checks with separate failure exits. Event-order quotients, history-response quotients, state abstractions, and action implementation correspondences are registered as four distinct map types.
2. **An exact track.** Under D5 and D6, macro kernels are well defined and finite-horizon state-path push-forward commutes for compatible control protocols (Section 3). Probabilities of persistence events are preserved only when the events are measurable through the aggregated paths.
3. **An approximate track.** Standard total-variation contraction and perturbation arguments provide a finite-horizon error budget with a separate term for target approximation (Section 4), together with explicit scope conditions.
4. **Executable failure witnesses.** Finite counterexamples CPIR-X1 to X8 kill eight wrong inferences, one per dropped premise, and the four model groups form a reproducible archive (Section 5).

The protocol deliberately relies on established lumpability, abstraction, and Markov-perturbation results; the proposed integration records how those ingredients apply to implementations, targets, and persistence events, and pairs them with finite failure witnesses. Any claim of an advantage over a specific existing framework requires a specified comparison; this paper does not assert an automatic gain in predictive or control capability.

2.5 What this paper does not claim

This is a preprint focused on the finite mathematical core of the DCS research program. This paper does not claim: a derivation of four-dimensional Lorentzian spacetime, quantum field theory, or the Standard Model; a composition theorem for open interconnected systems; a unified explanation of life, intelligence, or civilization; an intelligence capability lattice or collective agency certification; or empirical confirmation of any of its results in natural systems. The strict results are conditional results in finite, discrete-time, classical, registered task contracts; the new contributions are mainly typed stitching, certification structure, failure exits, and governance, not the reinvention of posets, lumpability, Markov contraction, or causal abstraction (Kemeny and Snell 1976; Buchholz 1994; Rubenstein et al. 2017; Beckers and Halpern 2019; Andrieux 2012).

The framework is designed to connect to the wider research programs that motivate cross-scale questions: free-energy and active-inference accounts of brain function treat perception and action as a single inferential economy (Friston 2010); autopoiesis theory asks what it means for an organization to persist as an identity (Maturana and Varela 1980); and reinforcement learning formalizes agency as the pursuit of future targets under action (Sutton and Barto 2018). This paper does not explain any of these programs; it provides a type-safe registration system in which their cross-scale claims can be stated, checked, and failed item by item.

2.6 Paper structure

Section 2 gives the finite model and the certification protocol, including the four certification obligations and the task contract. Section 3 gives the exact guarantees: per-action macro kernels, implementation-invariant macro action kernels, finite-horizon state-path commutation, and persistence-event preservation. Section 4 gives the approximate guarantees: the total-variation error budget, the contraction and linear bounds, and the target-approximation term. Section 5 reports executable instances and failure witnesses. Section 6 discusses related work, originality boundaries, and limitations. Section 7 concludes. Full proofs, the counterexample library, numerical models, and the proposition ledger are developed in the supplementary material.

3 Finite Model and Certification Protocol

3.1 Three causal duties and the typed discipline

The protocol separates three mutually non-identical causal duties. Structural reachability causality decides whether one event can reach another; dynamical realization decides how the dynamics is realized by generators, kernels, or history weightings; response-intervention causality decides what changes after an action, and requires action and target semantics (Pearl 2009; Rubenstein et al.

2017; Beckers and Halpern 2019). The intuition behind the typing discipline is close to the oldest lesson of structural realism: the structure that makes a system what it is should not be conflated with the implementation that realizes it (Ladyman 1998).

The consequence is a prohibition on untyped maps. Cross-scale maps are not a single coarse-graining but four type-distinct maps:

$$q^E : E \rightarrow \bar{E}, \quad \Phi^H : \mathcal{H} \rightarrow \mathcal{H}/\sim, \quad \tau : X \rightarrow Z, \quad \Gamma : \bar{\mathcal{A}} \rightrightarrows \mathcal{A}.$$

Map	Admissible input	Admissible output	Primary admission
q^E	events and event relations	event quotient	poset contract and structure preservation
Φ^H	full or truncated histories	history-response quotient	task response sufficiency
τ	micro states / state distributions	macro states / pushed-forward distributions	dynamical closure and representation error
Γ	macro actions	sets of micro implementations	implementability and implementation invariance

These types are not interchangeable: an event quotient is not defined by a state map, and Γ is a set-valued implementation correspondence that must not be silently converted into a single-valued function.

3.2 The controlled finite model

Definition 1 (finite classical controlled system). A finite classical controlled Markov system consists of a finite state space X , a finite action set $\mathcal{A}$, a finite horizon T , and row-stochastic kernels $P^a(x, \cdot)$ for each action a . Histories are elements of $\mathcal{H} = \bigcup_{t \leq T} (X \times \mathcal{A})^t \times X$.

Definition 2 (finite task contract). A finite task contract is

$$\Theta = (\Pi, F, T, \mathcal{E}, d, \delta),$$

where Π is the admissible strategy set, F the future target, T the finite horizon, $\mathcal{E}$ the environment, d the distance, and δ the componentwise error budget. The contract fixes, in one place, what may be done (strategies), what is being predicted (targets), how long the prediction runs (horizon), against what background (environment), measured in what distance, and with what error budget. All strict results in this paper are stated under finite instance task contracts; the general specification is not conflated with the finite instance.

The task contract is not decorative: each guarantee in Sections 3 and 4 is conditional on the corresponding component of the contract being registered. A target that is not registered cannot be claimed to be preserved; an error budget that is not registered cannot be claimed to be met.

3.3 Four certification obligations

The protocol registers four certification obligations separately.

Event-order quotient. An event quotient is a surjective map $q^E : E \twoheadrightarrow \bar{E}$ whose fiber graph has acyclic transitive closure (the AEQ condition), so that $(\bar{E}, \prec_q)$ is again a finite strict poset. This obligation certifies that the order structure survives aggregation. The AEQ condition is a separate premise, not an automatic property: the counterexample CPIR-X1 (Section 5.2) shows that an arbitrary partition of events can produce a cyclic macro relation, and CPIR-X2 shows that repairing one violated premise does not automatically restore the certificate.

History-response quotient. Define $h \sim_{\Theta} h'$ iff the structural response kernels $K_{t:T}^{\pi}(h, \cdot)$ and $K_{t:T}^{\pi}(h', \cdot)$ coincide for all $\pi \in \Pi$ and $0 \leq t \leq T$. The quotient $\Phi^H : \mathcal{H} \rightarrow \mathcal{H}/\sim_{\Theta}$ certifies that histories with identical registered responses are identified, and no more. Using structural response kernels avoids conditional-distribution versions at zero-probability histories.

State aggregation. A state abstraction $\tau : X \twoheadrightarrow Z$ with per-action strong blockability (Definition 3) certifies that macro dynamics is autonomous for each registered action.

Action implementation. An implementation correspondence $\Gamma : \bar{\mathcal{A}} \rightrightarrows \mathcal{A}$ with multi-implementation invariance (Definition 4) certifies that all micro implementations of a macro action have identical macro effects.

The separation is the content of the protocol: dynamical closure, target measurability, intervention consistency, and identity preservation are different obligations, and a single certificate that one holds carries no information about the others.

3.4 State aggregation and per-action macro kernels

Definition 3 (per-action strong blockability, D5). For each action a and macro state z' let $S^a(x, z') = \sum_{y \in B_{z'}} P^a(x, y)$, where $B_{z'}$ is the block of the partition corresponding to z' . The partition is strongly blockable for action a if $\tau(x) = \tau(x')$ implies $S^a(x, z') = S^a(x', z')$ for all z' . This is strong lumpability, applied action by action (Kemeny and Snell 1976; Buchholz 1994).

Lemma 1 (per-action macro kernels, conditional DCS result). Under D5, each micro action a induces a macro kernel $\bar{P}^a$ on Z , defined by $\bar{P}^a(z, z') = S^a(x, z')$ for $x \in B_z$. The kernel is indexed by the micro action a at this stage. The claim that all micro implementations of one macro action yield the same macro kernel requires multi-implementation invariance (D6) and is stated in Lemma 2.

3.5 Action implementation and macro action kernels

Definition 4 (multi-implementation invariance, D6). Let $\Gamma : \bar{\mathcal{A}} \rightrightarrows \mathcal{A}$ satisfy $\Gamma(\bar{a}) \neq \emptyset$. Multi-implementation invariance requires

$$S^a(x, z') = S^{a'}(x', z')$$

for any $\bar{a}$, any $a, a' \in \Gamma(\bar{a})$, $\tau(x) = \tau(x')$, and z' . This guarantees that all micro implementations of the same macro action have identical macro effects.

Lemma 2 (macro action kernels, conditional DCS result). Under D5 and D6, the macro action kernel

$$\bar{P}^{\bar{a}}(z, z') := S^a(x, z'), \quad x \in B_z, \quad a \in \Gamma(\bar{a}),$$

is independent of the choice of state representative and implementation representative, and is a stochastic kernel on Z for each $\bar{a}$.

The distinction between Lemma 1 and Lemma 2 is a recurring example of the protocol’s discipline: well-definedness at the level of individual micro actions (Lemma 1) is a weaker obligation than well-definedness at the level of macro actions with multiple implementations (Lemma 2). The counterexample CPIR-X4 (Section 5) shows that the second obligation can fail while the first holds.

3.6 Targets, persistence events, and memory

Targets are registered in the task contract as measurable functions on the relevant path space. Persistence indicators in this paper are generally history-dependent: the conditional persistence quantity G_T may depend on the whole history $X_{0:T}$, not only on the terminal state X_T .

This is why target measurability (D7) and persistence-event measurability (D8) are separate obligations with separate failure witnesses. A macro description may be dynamically closed while a target is not measurable through it, and identity-dependent persistence events may not be expressible at the macro level at all. In the approximate track, the distinction appears again: the total-variation bound applies to terminal-state targets, and persistence events inherit the bound only under explicit conditions (Section 4.3).

Memory augmentation (augmenting the state with the required history, $Z_t = (X_t, M_t)$) is a registered design choice: when the first-order Markov assumption fails, the augmented process must satisfy the same finite Markov and contraction premises before any of the guarantees apply. Augmentation is not a workaround that makes the premises automatic.

3.7 Closure is registered by grade

The protocol registers closure by grade rather than as a binary attribute:

- **D0-exact:** the macro kernel is exactly autonomous on the registered partition;
- **D1-approximate:** the macro kernel is approximately autonomous within the registered distance and error budget;
- **D2-memory-augmented:** the macro state requires augmented memory to recover task-relative closure.

D2 memory augmentation is written as $\tilde{Z}_t = (Z_t, M_t)$, with memory residual

$$\varepsilon_{\text{mem}} = I(F_{t:T}; H_t^- \mid Z_t, M_t, A_{t:T}).$$

When the first-order Markov assumption fails, one registers memory under D2 rather than hiding the memory or claiming memoryless closure. The grade appears in the task contract, so a certificate under D0 cannot be presented as covering a system that actually requires D2. This is another

instance of the protocol's central discipline: the obligation (here, memorylessness) is registered, checked, and failed explicitly rather than assumed.

4 Exact Guarantees

4.1 Finite-horizon state-path commutation

Lemma 3 (finite-horizon macro-micro state-path commutation, conditional DCS result). Let the initial distribution be μ , and let the macro strategy be admissible, randomized, and dependent on the macro history via strategy kernels

$$\bar{\pi}_t(\bar{a}_t \mid z_{0:t}, \bar{a}_{0:t-1}),$$

with micro implementations generated by compatible implementation kernels $\lambda_t(a_t \mid \bar{a}_t, h_t)$ whose support is contained in $\Gamma(\bar{a}_t)$. Then for every finite T ,

$$\tau_{\#}^{0:T} P_{\mathcal{C}}^{\mu, \pi} = P_{\mathcal{C}}^{\tau_{\#} \mu, \bar{\pi}}.$$

The commutation statement presupposes the implementation premises of Definitions 3 and 4, restated here because the result depends on them directly: (D5) each admissible micro action satisfies per-action strong blockability on the registered partition; (D6) the implementation correspondence is nonempty-covering and satisfies multi-implementation invariance. Without these two premises the macro kernel is not well defined and the commutation claim does not apply.

On the type of the equality. Here the push-forward $\tau_{\#}^{0:T}$ is defined on the state path space $X^{0:T}$, so the displayed equality is an equality of state-path distributions. The macro strategy $\bar{\pi}$ emits macro actions and the compatible implementation kernels λ_t emit micro actions $a_t \in \Gamma(\bar{a}_t)$; the formula itself does not assert a separate equality of joint state-action distributions on the unaugmented state path space. If a joint state-action statement is required, it must be defined on the augmented path space that includes the protocol-generated macro action records, with the joint push-forward $(\tau_{\#}^{0:T}, \text{id}_{\bar{A}^{0:T-1}})_{\#}$ used explicitly. Micro actions are not inverted to macro actions without the implementation relation, because implementation sets may overlap. A joint statement in that stronger form requires its own proof and is not claimed here. If a micro controller changes the macro action distribution directly from compressed information, without decomposing into the strategy-implementation protocol above, the induction step fails.

4.2 Target responses

If the target satisfies $F = \bar{F} \circ \tau^{0:T}$, then

$$\mathcal{L}(F \mid \mu, \pi) = \mathcal{L}(\bar{F} \mid \tau_{\#} \mu, \bar{\pi}).$$

Path commutation does not require D7; D7 is used only to pass from path-distribution equality to equal response distributions for a specified target. This is the point of the typing discipline: path

equality and target-response equality are different claims, and the protocol registers the bridge premise (target measurability) explicitly rather than silently assuming it.

4.3 The exact interface theorem

Theorem 1 (finite exact interface quotient, conditional DCS result). Suppose all of the following premises hold:

1. the event quotient q^E satisfies AEQ;
2. Φ^H is an exact history-response quotient under Θ ;
3. τ satisfies D5 for every admissible micro implementation;
4. Γ is nonempty-covering on all macro actions and satisfies D6;
5. targets, strategies, and initial distributions satisfy the Lemma 3 compatibility conditions.

Then the interface package

$$I_\Theta = (\bar{E}, \prec_q; Z, \bar{\mathcal{A}}, \{\bar{P}^{\bar{a}}\}; \mathcal{H}/\sim_\Theta)$$

has five properties: (a) the quotient event relation is a finite strict poset; (b) macro action kernels are well defined; (c) finite-horizon state-path push-forward commutes for all registered compatible strategies and initial distributions, with the target-measurability premise pushing the equality to target response distributions; (d) the history-response quotient preserves the responses registered in the task; (e) all four quotient types carry explicit lower-layer implementation fibers and effective domains.

Sketch of the argument. Property (a) follows from the AEQ acyclicity condition, which guarantees that the transitive closure of the fiber graph is again a finite strict poset. Property (b) is Lemma 2: D5 gives per-action well-definedness and D6 gives implementation independence. Property (c) is Lemma 3: the induction step uses per-action strong blockability to commute one time step, and the compatible strategy-implementation protocol to propagate the equality through the horizon. Property (d) is the definition of the history-response quotient, which identifies exactly the histories with identical registered responses. Property (e) holds by construction: each quotient type is accompanied by its lower-layer fibers and effective domain. Full proof in the supplementary material.

What cannot be concluded. Theorem 1 is a modular stitching theorem on finite objects: it proves that four quotient types and maps can coexist within a common task contract while preserving their respective types. It does not derive Markov dynamics from event posets alone, does not prove that arbitrary partitions are automatically lawful, does not prove that a certificate remains valid after repair of a violated premise, and does not prove that compression automatically generates capabilities.

4.4 Persistence-event preservation

Corollary 1 (persistence-event probability preservation under the quotient, conditional DCS result). Under the conditions of Theorem 1, suppose the persistence indicator satisfies $G_T = \bar{G}_T \circ \tau^{0:T}$; then

$$P_{\mathfrak{C}}^{\mu, \pi}(G_T = 1) = P_{\bar{\mathfrak{C}}}^{\tau \# \mu, \bar{\pi}}(\bar{G}_T = 1).$$

Here “descends to the quotient” means that G_T factors through $\tau^{0:T}$; it does not mean that the persistence probability numerically decreases. If two in-block micro histories have different identity judgments, the equation $G_T = \bar{G}_T \circ \tau^{0:T}$ fails and Corollary 1 cannot be invoked. The corollary does not apply to systems whose identity depends on compressed micro variables; the counterexample CIPR-X7 (Section 5) exhibits exactly this failure mode.

4.5 Premise necessity and failure exits

The protocol’s credibility rests on its failure exits: each premise of the exact track can be dropped, and the corresponding guarantee fails in a reproducible way. The eight counterexamples CIPR-X1 to X8 (Section 5.2) realize these failure modes in finite models.

Removed condition	Failing stage	Counterexample
AEQ acyclicity	macro event relation may cycle	CIPR-X1/X2
D5 strong blockability	macro kernel depends on in-block state	CIPR-X3
D6 implementation invariance	one macro action has multiple macro effects	CIPR-X4
D7 target measurability	paths commute but target depends on compressed variables	CIPR-X5
strategy-implementation compatibility	hidden micro information changes macro action distribution	CIPR-X5 extension
finite horizon / closure	latent variables may bias later memory	CIPR-X6
D8 persistence measurability	identity events cannot be expressed at the macro level	CIPR-X7
capability baseline and ablation	descriptive compression misread as capability growth	CIPR-X8

The table is read as a set of obligations, not as a set of excuses: a model that registers all premises has all guarantees; a model that drops any premise must either register the corresponding failure or provide a new proof for the stronger claim.

5 Approximate Guarantees

5.1 Total variation and the one-step joint defect

For probability distributions μ, ν on a finite set Z ,

$$d_{\text{TV}}(\mu, \nu) = \frac{1}{2} \sum_{z \in Z} |\mu(z) - \nu(z)|.$$

The variational form gives $|\mathbb{E}_\mu f - \mathbb{E}_\nu f| \leq d_{\text{TV}}(\mu, \nu)$ for $f : Z \rightarrow [0, 1]$, so bounds on total variation translate directly into bounds on the discrepancy of expected targets.

Define the stepwise defect between the pushed-forward micro kernel and the macro kernel:

$$\delta_t = \sup_{x \in X} d_{\text{TV}}(\tau_{\#} P_t(x, \cdot), \bar{P}_t(\tau x, \cdot)).$$

δ_t simultaneously contains state-abstraction error and macro-kernel error; it is positive whenever the macro-micro map does not commute exactly. The supremum is over micro states, so the defect is a worst-case per-state measure, not an average.

5.2 Contraction and the finite-horizon bound

For a Markov kernel $\bar{P}_t$ on Z , the Dobrushin coefficient is

$$\alpha_t = \sup_{z, z' \in Z} d_{\text{TV}}(\bar{P}_t(z, \cdot), \bar{P}_t(z', \cdot)),$$

and $d_{\text{TV}}(\mu \bar{P}_t, \nu \bar{P}_t) \leq \alpha_t d_{\text{TV}}(\mu, \nu)$ is a standard Markov contraction tool (Andrieux 2012). Writing $e_t = d_{\text{TV}}(\tau_{\#} \mu_t, \bar{\mu}_t)$ for the macro-micro distribution error at time t , the triangle inequality, the distribution-defect lifting lemma, and contraction give the one-step recursion

$$e_{t+1} \leq \delta_t + \alpha_t e_t.$$

The intuition: the new error is at most the one-step coupling defect plus the residual of the old error after the macro kernel contracts it. Recursing gives the telescoping bound

$$e_T \leq e_0 \prod_{j=0}^{T-1} \alpha_j + \sum_{k=0}^{T-1} \delta_k \prod_{j=k+1}^{T-1} \alpha_j,$$

which decomposes the total error into the decayed initial error and the weighted accumulation of stepwise defects, with weights determined by subsequent contraction coefficients.

Theorem 2 (finite approximate error theorem, conditional DCS result). If $\alpha_t \leq \alpha < 1$ and $\delta_t \leq \delta$, then

$$e_T \leq \alpha^T e_0 + \delta \frac{1 - \alpha^T}{1 - \alpha}.$$

If only $\alpha = 1$ is available, only the linear finite-horizon bound remains: $e_T \leq e_0 + T\delta$.

Wording and scope. The initial discrepancy contracts geometrically under contraction, while continuing nonzero defects accumulate to a bounded error envelope; without contraction only the

linear bound remains. The theorem applies to finite, discrete-time, classical Markov models under total-variation protocols; it does not apply to continuous, quantum, infinite-state, or strongly non-Markovian settings. The distinction between the contractive bound and the linear bound is not a decoration: the numerical execution certificate in Section 5.3 freezes both regimes.

5.3 Targets and persistence events

If the macro target $f : Z \rightarrow [0, 1]$ is used, the variational form gives

$$|\mathbb{E}_{\tau_{\#}\mu_T}f - \mathbb{E}_{\bar{\mu}_T}f| \leq e_T.$$

If the micro target F is only approximately measurable, $\sup_x |F(x) - f(\tau x)| \leq \varepsilon_F$, then

$$|\mathbb{E}_{\mu_T}F - \mathbb{E}_{\bar{\mu}_T}f| \leq \varepsilon_F + e_T.$$

The target-approximation error ε_F must be registered separately from the dynamical defect δ_t and must not be double-counted. If micro states in the same macro block have different targets or identities, ε_F can reach 1, and dynamical closure cannot substitute for a target-measurability certificate; the counterexamples CPIR-X5 and CPIR-X7 (Section 5) exhibit this regime.

Persistence events. The bound above is stated for terminal-state targets $F : X \rightarrow [0, 1]$ registered in the task contract. Persistence indicators in this paper are generally history-dependent: the conditional persistence quantity G_T may depend on the whole history $X_{0:T}$, not only on the terminal state X_T . The T2 bound therefore inherits to a persistence event only in one of the following cases: (i) the persistence event is a terminal-state function of X_T ; or (ii) the required history has been encoded into an augmented state $Z_t = (X_t, M_t)$ under memory augmentation, and the augmented process satisfies the same finite Markov and contraction premises. Extending the bound to path-dependent persistence probabilities in general requires an augmented-state or path-space measure argument and is future work; Theorem 2 does not claim automatic inheritance for all persistence events.

5.4 The error budget as a contract component

The approximate track returns an error budget with three named terms: the decayed initial error (controllable by initialization and contraction strength), the accumulated defect envelope (controllable by the coupling defect and the horizon), and the target-approximation term (controllable by target measurability). Registering the budget in the task contract makes the claim checkable: a model that reports a numerical error must identify which terms it computed and under which regime (contractive or linear). The protocol does not assert that the bound is tight in general; it asserts that the bound is explicit, registered, and machine-verifiable on the frozen instances.

6 Executable Instances and Failure Witnesses

All numerical results in this paper are executable certificates within specified finite models. The code, fixed configurations, expected outputs, tests, and reproduction instructions form the reproduction archive at <https://github.com/weijierry/dcs-finite-certification> (commit 2c15258, 2026-09-10). The four model groups are `cpir_exact`, `cpir_approximate`, `selective_compression_validation`, and `cross_scale_interface`; each is deterministic given its registered inputs, and the regression tests compare against frozen baselines, with a 1e-12 tolerance where the numerics depend on the BLAS/LAPACK build.

What validation means for a certification framework. A framework whose main export is certification is only as credible as its failure exits: the same machinery that admits an interface must be able to reject one, and the rejection must be reproducible. The finite models in this paper play that role: they are positive examples, mandatory counterexamples, boundary samples, and stress tests, not empirical evidence about nature.

6.1 Exact positive example

The registered positive instance (`models/cpir_exact`) has a structure in which the registered partition has zero lumpability defect, zero target-measurability defect, and zero persistence-measurability defect, is implementation-invariant, and yields a well-defined macro kernel:

```
implementation_invariant: true
lumpability_defect: 0.0
macro_kernel: [[0.8, 0.2], [0.3, 0.7]]
persistence_measurability_defect: 0.0
target_measurability_defect: 0.0
```

Together with the counterexamples below, this positive example constitutes the machine regression gate for Theorem 1 and Corollary 1. Passing the tests unlocks the machine regression gate; the theorem identity is still jointly confirmed by the main-text proof, source mapping, and human review.

6.2 Failure witnesses Cpir-X1 to X8

Eight counterexamples kill the following wrong inferences:

ID	Inference killed	Key failing field
X1	arbitrary partitions automatically yield lawful poset quotients	macro_cycle: true
X2	certificates remain valid after repair of a violated premise	requires_recertification: true
X3	arbitrary partitions automatically yield Markov macro kernels	lumpability_defect: 0.8

ID	Inference killed	Key failing field
X4	macro actions naturally have unique micro-implementation effects	implementation_invariant: false
X5	dynamical closure automatically guarantees task sufficiency	target_measurability_defect: 1.0
X6	one-step fits automatically guarantee multi-step memorylessness	two_step_equal: false
X7	dynamical quotients automatically preserve persistence identity	persistence_measurability_defect: 1.0
X8	descriptive compression automatically generates new capabilities	new_reachable_capability: false

Two of these witnesses carry the protocol’s central message. CPIR-X4 shows that one macro action can have multiple macro effects when D6 fails, which is exactly the gap between Lemma 1 and Lemma 2. CPIR-X5 shows that distribution closure does not certify a non-macro-measurable target: the macro distribution error can be zero while a hidden target reaches maximal error. The complete “killed inference, minimal construction, failing field, repair exit” table for each counterexample appears in the supplementary material.

6.3 T2 numerical execution certificates

`models/cpir_approximate` checks, time step by time step, that for the finite instance $\alpha = 0.5$, $\delta = 0.02$ the actual total-variation error does not exceed the geometric bound:

Horizon T	Actual error	Theoretical bound
1	0.02	0.02
4	0.0025	0.0375
8	0.000156	0.039844
12	0.000010	0.039990

Two boundary samples are also frozen: with $\alpha = 1$ only the linear finite-horizon bound is available (0.16 at $T = 8$); and with zero macro distribution error, a hidden, non-macro-measurable target can still reach maximal error. These samples verify implementation and counterexample responsibility; they are not empirical confirmation.

6.4 Non-reversible stress test

The non-reversible circulation family (`models/selective_compression_validation`) shows, under a scan of heterogeneity h , high directional alignment coexisting with joint admission failure. Regis-

tered thresholds: response distortion at most 0.01, closure defect at most 0.005, two-step memory residual at most 0.005, dynamics-response direction cosine at least 0.99.

h	Response distortion	Closure defect	Memory residual	Direction cosine
0.0	0.0	0.0	0.0	1.0
0.1	0.00860	0.00400	0.00323	0.99998
0.2	0.01730	0.00800	0.00649	0.99993
0.8	0.07415	0.03200	0.02633	0.99889

The joint admission plateau endpoint is $h = 0.1$; at $h = 0.2$ all three errors cross the thresholds while the direction cosine remains as high as 0.99993. Allowed conclusion: high directional alignment can coexist with joint interface admission failure. Forbidden conclusion: a causal plateau in a real natural system has been empirically discovered.

6.5 Joint admission prototype

`models/cross_scale_interface` exhaustively enumerates all 15 partitions of the surviving states of a five-state, three-action model. In the symmetric model the registered partition $\{A0, A1\}|\{B0, B1\}|\{D\}$ simultaneously has zero within-class response distortion, zero closure defect, zero two-step memory residual, zero intervention-commutation error, and zero persistence distortion. At within-block heterogeneity 0.25, response, closure, memory, and persistence distortions turn positive while intervention commutation remains zero. This single instance shows that intervention commutation is not equivalent to the other admission conditions in this model; it does not by itself establish that all certification axes are pairwise logically independent, which would require separately constructed distinguishing witnesses for each pair of axes. The raw Pareto frontier is fully preserved, to exhibit that “one-block collapse” may remain on the frontier because it minimizes interface cost: Pareto membership itself does not constitute effective-layer certification.

6.6 Regression gates

The reproduction archive carries mandatory regression gates before freezing: the registered tests for closure, action, and counterexample obligations must pass, and any unconditional occurrence in the main text of “closure implies stability,” “stationarity implies attraction,” “organization implies life,” or “order implies manifold” must fail the audit. The archive’s test set is reported separately from the full project suite: the release contains four model test files (one per group), and the counts are not merged with historical full-project totals. Passing the release tests unlocks the machine regression gate; it does not by itself certify the mathematics, which remains jointly confirmed by the main-text proofs, source mapping, and human review.

7 Related Work and Limitations

7.1 Comparison method

Each obligation in the protocol is compared with the established method that addresses it. For each item, the comparison records what the existing method already checks, what this protocol additionally requires, and what reproducible difference the additional check produces. Conceptual juxtaposition is not originality, renaming is not derivation, and cross-domain coverage is not evidence. The comparison distinguishes “not attempted by an existing method” from “an error of an existing method”: standard lumpability never guaranteed that arbitrary micro identity labels are macro-measurable, and no claim here asserts otherwise.

7.2 Established results and their roles

Established result	Existing scope / conditions	Role in this paper	Difference to be checked
Finite posets, Hasse diagrams, linear extensions (Bombelli et al. 1987; Norris 1997)	finite strict posets	AEQ contract for event quotients	infinite poset topology and measures; manifold embedding
Finite generators, irreducible CTMC, KL contraction (Norris 1997; Kemeny and Snell 1976)	finite states, conservative, registered time	typed integration of stochastic semigroups into the contract	non-normal generator spectra; growing-system limits
Strong lumpability (Kemeny and Snell 1976; Buchholz 1994)	fixed partition, strong blockability	per-action strong blockability (Definition 3, Lemma 1)	global spectral gap does not guarantee partition blockability
Causal consistency / causal abstraction (Rubenstein et al. 2017; Beckers and Halpern 2019; Xia and Bareinboim 2025)	macro-micro intervention commutation	implementation invariance and task responses (Definition 4, Lemma 2)	latent variables and selection bias make targets non-identifiable
Computational mechanics / predictive equivalence (Crutchfield and Young 1989; Shalizi and Crutchfield 2001)	predictively sufficient states	response targets separated from predictive compression	transferability under mechanism change
Information bottleneck (Tishby et al. 1999)	mutual-information compression-relevance tradeoff	predictive vs causal compression separated	intervention semantics not preserved by mutual information

Established result	Existing scope / conditions	Role in this paper	Difference to be checked
Mori-Zwanzig projection (Zwanzig 1961; Zwanzig et al. 1972; Mori 1965)	projection-memory structure	memory residual registered as an obligation	strongly coupled coarse-graining; non-Markov closure
Finite-time coarse-graining error bounds (Andrieux 2012)	explicit kernels, distances, mapping assumptions	Theorem 2 error budget	this paper’s TV protocol differs in conditions; no claim of being inherently stronger

Two boundary literatures deserve explicit mention. First, the physics bridging direction concerns reconstructions of spacetime and symmetry from causal and geometric data, including causal set theory (Sorkin 2005) and order- and number-based reconstruction programs (Braun 2026; Braun and Sämann 2025); none of these is implied by the finite results of this paper, and the paper does not claim otherwise. Second, non-Markovianity is increasingly a measurable, controlled property: process-tensor methods characterize and control non-Markovian quantum processes experimentally (White et al. 2020), which is exactly why memory augmentation is a registered design choice and not a workaround.

7.3 Originality boundaries

The mathematical ingredients are established: strong lumpability (Kemeny and Snell 1976; Buchholz 1994), causal consistency of structural equation models (Rubenstein et al. 2017), and finite-time coarse-graining error bounds for hidden Markov dynamics (Andrieux 2012). The originality of this paper should be stated only as: placing these mature results into one typed task contract, requiring events, histories, states, interventions, targets, and persistence events to be certified item by item, and providing a jointly failing interface package with reproducible failure witnesses. It must not be claimed that this paper originated posets, lumpability, Markov contraction, causal abstraction, or the information bottleneck. When citing Mori-Zwanzig, the correction must be retained: only the projection-memory structure is inherited, not the uncorrected nonlinear transport conclusion (Zwanzig 1961; Zwanzig et al. 1972).

7.4 Relationship to the wider cognitive, life, and AI programs

The free-energy principle proposes that any adaptive system can be described as minimizing variational free energy, unifying perception, action, and learning under one normative account (Friston 2010); active inference gives that principle a process-level implementation. This paper does not compete with that program; it is strictly more modest: its task contracts provide a registration format in which claims about cross-scale causal interfaces (including, in principle, those made by free-energy and active-inference models) can be checked axis by axis. Similarly, autopoiesis theory and the theory of major transitions define persistence and individuation at the level of living organization (Maturana and Varela 1980; Maynard Smith and Szathmary 1995), and reinforcement learning provides the operational definition of a task: an agent, a set of actions, a reward signal,

and a horizon (Sutton and Barto 2018). Each of these traditions supplies one facet of the cross-scale question; none of them, on its own, provides the typed, failure-aware certification structure proposed here. Structural realism supplies the philosophical warning that motivates the enterprise: structure and implementation must be kept distinct if a theory is to survive theory change (Ladyman 1998).

7.5 Limitations

All strict results are confined to finite, discrete-time, classical Markov models under total-variation protocols. Infinite poset topology and measures, identifiability of native weights, non-explosion and local limits of growing systems, spectral analysis of non-normal generators, strongly coupled coarse-graining, non-Markov closure, and renormalization-group fixed points remain open. State spaces grow super-exponentially with node count; exhaustive partition search does not scale; metastable lifetimes can far exceed simulation windows. This paper has no general scale-existence theorem, and it has not proved that cross-domain scale indicators admit a common normalization.

The numerical examples are synthetic finite-model computations; no empirical confirmation in natural systems is claimed. No general compositional guarantee for open interconnected systems is provided. Independent reproduction has not yet been performed by an independent party; the reproduction archive enables but does not certify it. In particular, the protocol does not assert that its bounds are tight, that its counterexamples exhaust all failure modes, or that a certificate valid under the registered premises remains valid after the premises change.

7.6 Conjectures and termination conditions

Conditional conjectures and future work are registered separately from strict results. In particular, extending the approximate bound to general path-dependent persistence events, establishing the operational value of the four-map integration through specific comparisons, and identifying cross-sample-stable scale-organization frontiers remain open items and are not claimed here. The framework carries its own termination condition: if the typed certification structure cannot produce fewer errors or stronger predictions than existing tools, or if its checks show no stable gain across models, the program should be reduced to a methodological framework rather than expanded into a unified theory. Termination, contraction, or downgrade is also a scientific result.

8 Conclusion

8.1 What is strictly established

Under finite task contracts, this paper establishes the following.

1. **A typed decomposition of certification obligations.** State aggregation (D5), action implementation (D6), target measurability (D7), and persistence-event measurability (D8) are separate premises with separate failure exits; event-order quotients, history-response quotients, state abstractions, and action implementation correspondences are four distinct map types that must not be conflated.

2. **An exact track.** Under D5 and D6, per-action macro kernels (Lemma 1) and implementation-invariant macro action kernels (Lemma 2) are well defined; for compatible control protocols, finite-horizon state-path push-forward commutes (Lemma 3); the exact interface theorem (Theorem 1) assembles the four quotient types into one interface package under five explicit premises; and persistence probabilities are preserved exactly when the persistence event is quotient-measurable (Corollary 1).
3. **An approximate track.** The total-variation error budget (Theorem 2) separates the decayed initial error, the accumulated defect envelope, and the target-approximation term, with explicit conditions under which the contractive or linear bound applies, and explicit conditions under which persistence events inherit the bound.
4. **Executable failure witnesses.** Eight finite counterexamples (CPIR-X1 to X8) kill eight wrong inferences, one per dropped premise, and four model groups form a reproducible archive with frozen baselines and regression tests.

The mathematical ingredients are established results; the contribution is the explicit organization of certification obligations and their finite failure witnesses.

8.2 What is only provided

The numerical results are executable certificates and mandatory counterexamples within finite models. They do not constitute empirical confirmation in natural systems. The paper does not claim general compositional guarantees for open systems, nor a derivation of spacetime, life, intelligence, or civilization, nor an automatic gain in predictive or control capability. Any claim of an advantage over a specific existing framework requires a specified comparison that this paper does not assert in general terms.

8.3 What remains open

The following items are registered as open and are not claimed here:

- general path-dependent persistence bounds (extending Theorem 2 beyond terminal-state or augmented-state conditions);
- the operational value of the four-map integration relative to specific existing frameworks;
- cross-sample-stable scale-organization frontiers and their characterization;
- independent reproduction by an independent party with a documented record;
- identifiability of native model weights, spectral analysis of non-normal generators, and strongly coupled coarse-graining;
- any claim concerning infinite systems, open interconnected systems, or natural-system validity.

8.4 The boundary

This protocol forms a bounded mathematical component of the DCS research program. Its claims are limited to specified finite models, and its statements distinguish standard results, conditional DCS results, and conjectures. The preprint has not been peer-reviewed; the reproduction archive invites scrutiny and correction, and the framework’s own termination conditions apply to the program as a whole.

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